

Module – 4

Boolean Algebra and Logic Circuits

- Convert the following numbers to its equivalent numbers and show the steps:
 - $(10110001101011.111100000)_2 = (?)_8$
 - $(10110001101011.11110010)_2 = (?)_{16}$
- Convert Decimal to Binary: (i) 0.6875 (ii) 0.513
- Convert Binary to Decimal:
 - 110111 (ii) 10101010
- Convert:
 - $(110.1101)_2 = (?)_{10}$
 - $(CAD.BF)_{16} = (?)_{10}$
 - $(623)_8 = (?)_{10} = (?)_2 = (?)_{16}$
 - $0.6875_{(10)} = ?_{(2)}$
 - Compute the 9's complement of $25.639_{(10)}$
 - Compute the 1's complement of $11101.0110_{(2)}$
- Subtract the following using 10's complement:
 - $(72532 - 3250)_{10}$
 - $(3250 - 72532)_{10}$
- Perform subtraction on the given numbers using 9's complement method:
 - $4,637 - 2,579$ (b) $125 - 1,800$
- Perform subtraction on the given numbers using 10's complement method:
 - $2,043 - 4,361$ (b) $1,631 - 745$
- Perform subtraction on the given binary numbers using 1's complement method:
 - $10011 - 10010$ (b) $100010 - 100110$
- Perform subtraction on the given binary numbers using 2's complement method:
 - $1001 - 110101$ (b) $101000 - 10101$
- Subtract using $(r - 1)$'s complement method **(MQP '22 – 6M)**
 - $4456_{(10)} - 34234_{(10)}$
 Subtract using r 's complement method
 - $1010100_{(2)} - 1000100_{(2)}$
- State and prove De Morgan's theorem with its truth table.
- Using basic Boolean theorems, prove
 - $(x + y)(x + z) = x + yz$
 - $xy + xz + y\bar{z} = xz + y\bar{z}$
 - $x'y'z + x'yz + xy'$
 - $xy + x'z + yz$
 - $(x + y)(x' + z)(y + z)$
- Minimize the following function
 - $F(x, y, z) = xy + x'z + yz$

13. Express the Boolean function
 - i) $F = A + \overline{B}C$ in a sum of minterms form
 - ii) $F = xy + \overline{x}z$ in a product of maxterms form
14. Explain all the logic gates with the symbols and truth tables.
15. Implement half adder using basic gates.
16. Design a full adder using two half adders and an OR-gate.

Module – 5

Introduction to Transducers

1. Explain the working of the potentiometric resistive transducer.
2. With neat diagram, explain the operation of a piezoelectric transducer.
3. Explain the working principle and applications of piezoelectric transducer.

Communications

1. Explain the various blocks involved in communication block diagram.
2. Define modulation. Explain the need for modulation.